A120589 concise calculus certificate

VERIFIED

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Definition and first terms

$A(x) = A_{\text{parent}}(x)^2, \quad A_{\text{parent}}(x) = A_{120588}(x)$, with the origin-normalized solution.

Normalization/grammar: `ordered forest of 2 parent A120588 typogeometries`.

$a(0), \ldots, a(7) = 1, 2, 3, 6, 15, 42, 126, 396.$

Kernel used throughout: $\rho(u) = u\left(1 - 1\,u^1\right)$, so $x = \rho(T(x))$.

Typogeometry

`{0,1}x1` `{0,{1,1}}x1` `{0,{1,{1,1}}}x1` `{{{1,1},1},0}x1`

forest slots are explicit; 0 denotes an empty slot. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2 : 1$.

Coefficient and contour extraction

$$p(n) = [x^n]A_{\text{parent}}(x),$$

$$a(n) = \sum_{\substack{n_1,\ldots,n_2 \geq 0 \\ n_1+\cdots+n_2=n}} \prod_{j=1}^{2} p(n_j).$$

$$[x^n]A(x)^2 = \frac{2}{2\pi i n} \oint_\gamma \frac{(1+u)^1\,du}{u^n D(u)^n}.$$

Recurrence

$\sum_{r=0}^{2} P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$P_0(n) = 0$$
$$P_1(n) = -4\,n - 2$$
$$P_2(n) = n + 2$$

Differential equation

$$(2\,x)\,A(x) + \left(-4\,x^2 + x\right)A'(x) = 2\,x^2 + 4\,x.$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/(-u^2+u)^1$.

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))$.

Matrix dimensions: 4×4; remainder 2×3, rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.